

TrES-1b

R-2073

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_120330 · created 2026-08-09T23:25:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456016.89125 +/- 0.00086 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.012
Transit depth (Rp/Rs)^2	2.34 +/- 0.36 [%]
Orbital Inclination (inc)	87.18 +/- 1.02
Airmass coefficient 1 (a1)	0.99 +/- 0.012
Airmass coefficient 2 (a2)	0.0044 +/- 0.0018
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	#2 - [161, 348]
Optimal Aperture	4.47
Optimal Annulus	12.25
Transit Duration (day)	0.0929 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.012 vs 0.1358 (deviation 1.23σ)
Duration fit vs published	0.0929 d vs 0.1042 d (deviation 1.01σ)
Mid-time O–C	-2.1 min
Depth SNR	10.9
Residual scatter	1.25 %

Light curve

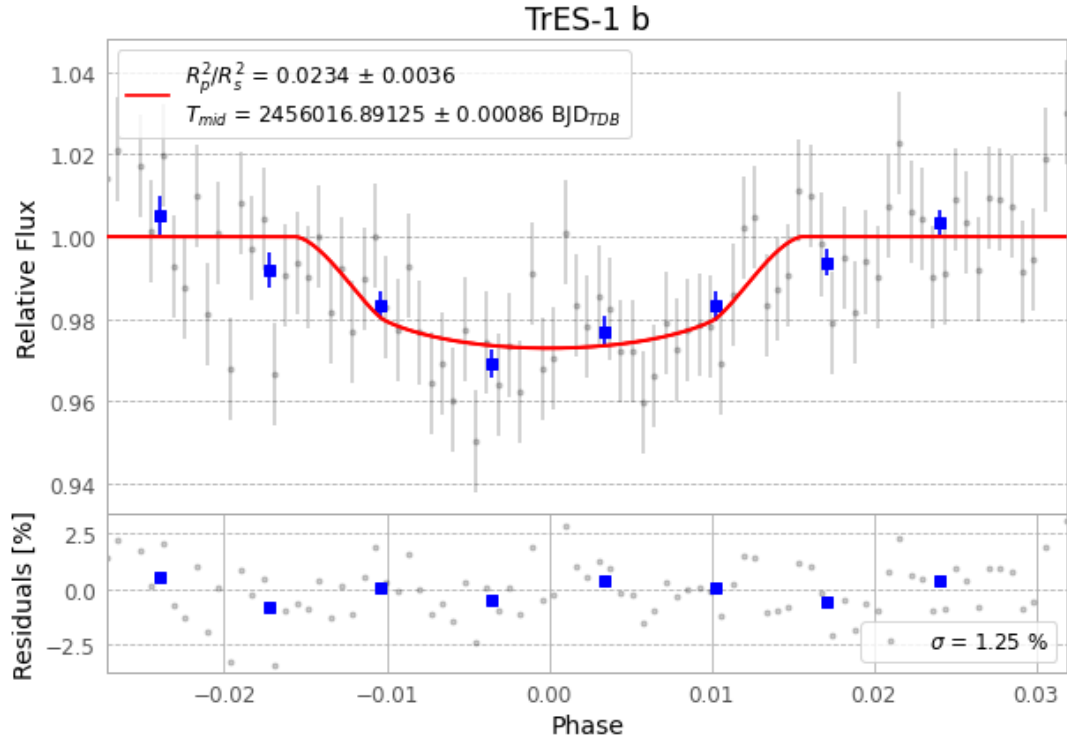


Figure 1 — FinalLightCurve_TrES-1 b_2012-03-30.png (SHA-256 c7ac0e67fd26c83c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [185, 115], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1145b30aebc9f0c1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

88 FITS frames (58 MB), TRES-1120330072412.FITS → TRES-1120330114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-120330.log (d2c96b9a0336...), FinalLightCurve_TrES-1 b_2012-03-30.png (c7ac0e67fd26...), FinalLightCurve_TrES-1 b_2012-03-30.csv (7bb039c8a6df...)

Compute resources

Wall time 172.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:25Z.